



Geopolitical Aviation Risk & Impact System

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Project Information

- The Context: Global aviation faces massive disruptions when regional conflicts occur (e.g., the Middle East corridor).
- The Goal: To move beyond "guesses" and use data to calculate exactly how much money and time are lost when airspace closes.
- Data Sources: Integrated 7+ datasets including Airline Losses, Flight Reroutes, Airport Disruptions, and Conflict Event Logs.



Count the number of conflict_events grouped by their severity level (High, Critical, etc.)

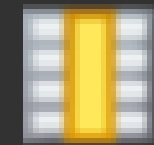
```
SELECT
    severity, COUNT(*) AS Total_Count
FROM
    conflict_events
GROUP BY severity;
```

Result Grid   Filter Rows:			
	severity	Total_Count	
☐	High	9	
☐	Medium	22	
☐	Very High	3	
☐	Critical	7	
☐	Low	9	

Calculate the total estimated_loss_usd for all airlines in the airline_losses table.

```
SELECT
    SUM(estimated_loss_usd) AS Total_loss
FROM
    airline_losses;
```

Result Grid



Total_loss

9990020000



List all unique airline names that appear in the airline_losses dataset.

```
SELECT DISTINCT
*
FROM
airline_losses
ORDER BY airline ASC;
```

Result Grid   Filter Rows: Export: 

	airline	country	airline_type	estimated_loss_usd	cancellations_count	reroutes_count	revenue_loss_pct	region	
	Air Arabia	UAE	Low Cost	190120000	114	45	43.5	Middle East	
	Air France	France	Flag Carrier	89670000	30	10	11.4	Europe	
	Air India	India	Flag Carrier	67960000	108	56	10.3	South Asia	
	American Airlines	USA	Flag Carrier	34580000	21	30	4.8	North America	
	British Airways	UK	Flag Carrier	42710000	51	28	4.5	Europe	
	Cargolux	Luxembourg	Cargo	44440000	18	47	12.3	Europe	
	Cathay Pacific	Hong Kong	Flag Carrier	30530000	9	26	4.9	Asia Pacific	
	Delta Air Lines	USA	Flag Carrier	80540000	47	45	8.3	North America	
	DHL Aviation	Germany	Cargo	18960000	9	86	12.9	Europe	
	EgyptAir	Egypt	Flag Carrier	99760000	65	62	15.4	Africa	
	Emirates	UAE	Flag Carrier	1192640000	187	130	20.5	Middle East	
	Emirates SkyCa...	UAE	Cargo	582080000	174	31	48.1	Middle East	
	Ethiopian Airlines	Ethiopia	Flag Carrier	34390000	17	40	4	Africa	
	Etihad Airways	UAE	Flag Carrier	324960000	118	106	34.5	Middle East	
	FedEx Express	USA	Cargo	102640000	38	86	14.2	North America	
	flydubai	UAE	Low Cost	924590000	71	94	50.6	Middle East	
	Gulf Air	Bahrain	Flag Carrier	421170000	114	64	28.8	Middle East	
	IndiGo	India	Low Cost	131380000	22	71	24.5	South Asia	
	Iran Air	Iran	Flag Carrier	297950000	193	57	44.5	Middle East	
	Jazeera Airways	Kuwait	Low Cost	309490000	137	35	40.4	Middle East	

Find all flight reroutes where the delay_hours exceeded 4 hours.

```
2 SELECT
3     *
4 FROM
5     flight_reroutes
6 WHERE
7     delay_hours > 4;
```

airline	flight_number	origin	destination	original_route	new_route	original_distance_km	new_distance_km	additional_distance_km	extra_fuel_cost_u...	delay_hours
Ethiopian Airlines	ET4033	Dubai	Mumbai	Via Iran/Iraq corridor	Via Egypt–East Africa southern route	11415	13536	2121	13194.93	5.5
Saudi Arabian Airlines	SA3737	Istanbul	Delhi	Via Iran/Iraq corridor	Via Egypt–East Africa southern route	3274	5555	2281	9082.97	4.4
flydubai	FL5718	Istanbul	Karachi	Direct Tehran overfly	Via Ashgabat–Almaty route	11556	12484	928	5471.51	4.2
British Airways	BR152	Dubai	Amman	Direct Tehran overfly	Via Ashgabat–Almaty route	8959	10615	1656	6360.78	4.7
flydubai	FL5920	Dubai	Istanbul	Direct Middle East overfly	Via Central Asian detour	5361	6804	1443	7996.65	5.7
Air Arabia	AI4779	London	Doha	Via Iran/Iraq corridor	Via Egypt–East Africa southern route	8469	11211	2742	13227.79	5.0
Etihad Airways	ET6117	Cairo	Kuwait City	Via Persian Gulf	Via Indian Ocean southern route	4248	7725	3477	15602.36	5.4
Ethiopian Airlines	ET6289	Riyadh	Cairo	Via Iran/Iraq corridor	Via Egypt–East Africa southern route	8084	9898	1814	9402.62	4.4
Jazeera Airways	JA2514	London	Dubai	Direct Tehran overfly	Via Ashgabat–Almaty route	5473	7428	1955	8037.33	5.8
Ethiopian Airlines	ET7975	Riyadh	London	Direct Tehran overfly	Via Ashgabat–Almaty route	6759	7718	959	5374.54	4.8
IndiGo	IN8637	Istanbul	Dubai	Direct Middle East overfly	Via Central Asian detour	9234	10404	1170	4140.07	4.5
Lufthansa	LU3916	Dubai	Frankfurt	Via Baghdad FIR	Via Ankara–Baku corridor	5205	6444	1239	5393.63	4.8
United Airlines	UN249	Beirut	Paris	Direct Middle East overfly	Via Central Asian detour	4711	6738	2027	9647.84	4.7
DHL Aviation	DH2120	Dubai	New York	Direct Tehran overfly	Via Ashgabat–Almaty route	10152	11990	1838	8217.23	5.9
Pakistan Internation...	PA346	Karachi	Dubai	Direct Middle East overfly	Via Central Asian detour	3191	5058	1867	8527.91	4.7
Lufthansa	LU8629	New Y...	Dubai	Via Strait of Hormuz	Via Red Sea–Suez route	10722	11811	1089	4947.27	5.7
Etihad Airways	ET7151	Dubai	Karachi	Via Iran/Iraq corridor	Via Turkey–Kazakhstan northern ro...	3931	5136	1205	5455.95	4.8
Kuwait Airways	KU5166	Frankf...	Dubai	Via Iran/Iraq corridor	Via Turkey–Kazakhstan northern ro...	5248	6134	886	3630.51	4.9
Pakistan Internation...	PA8070	Istanbul	Delhi	Direct Gulf route	Via Jordan–Egypt alternate	3661	4184	523	2513.54	6.0
Ethiopian Airlines	ET2226	London	Singapore	Direct Gulf route	Via Jordan–Egypt alternate	5244	5890	646	2698.87	4.6


```
3 SELECT
4     *
5 FROM
6     Aviation.airport_disruptions
7 WHERE
8     flights_affected > 100
9 ORDER BY duration_hours;
```

List airports from airport_disruptions where the flights_affected are more than 100, sorted by duration_hours.



airport_name	iata_code	country	region	disruption_type	severity_level	flights_affected	duration_hou...	date
King Abdulaziz International Airport	JED	Saudi Arabia	Middle East	Security Lockdown	High	289	2.8	2026-03-10
Indira Gandhi International Airport	DEL	India	South Asia	Departure Delays	High	241	3.2	2026-04-08
Beirut-Rafic Hariri International Airport	BEY	Lebanon	Middle East	Security Lockdown	Low	200	4.9	2026-03-30
Chhatrapati Shivaji Maharaj International Airport	BOM	India	South Asia	GPS Interference Advisory	High	245	5.6	2026-04-14
London Heathrow Airport	LHR	UK	Europe	GPS Interference Advisory	Low	222	6.4	2026-03-25
Kuwait International Airport	KWI	Kuwait	Middle East	Ground Stop	High	123	6.6	2026-04-16
Charles de Gaulle Airport	CDG	France	Europe	Ground Stop	Low	248	6.8	2026-04-17
Dubai International Airport	DXB	UAE	Middle East	Departure Delays	High	138	7.2	2026-03-30
Istanbul Airport	IST	Turkey	Middle East	Flight Cancellations	High	295	10.3	2026-03-11
Damascus International Airport	DAM	Syria	Middle East	Security Lockdown	Severe	187	12	2026-03-21
Istanbul Airport	IST	Turkey	Middle East	Arrival Diversions	Severe	148	12	2026-03-22
Damascus International Airport	DAM	Syria	Middle East	Ground Stop	Severe	225	12	2026-03-22
Mehrabad International Airport	THR	Iran	Middle East	Security Lockdown	Severe	139	12	2026-04-08
Beirut-Rafic Hariri International Airport	BEY	Lebanon	Middle East	ATC Flow Restriction	Severe	132	12	2026-04-09
Basra International Airport	BSR	Iraq	Middle East	Full Closure	Severe	198	12	2026-04-09
King Abdulaziz International Airport	JED	Saudi Arabia	Middle East	Flight Cancellations	High	282	12.2	2026-03-15
Ben Gurion Airport	TLV	Israel	Middle East	ATC Flow Restriction	High	131	14.1	2026-04-03
Cairo International Airport	CAI	Egypt	North Africa	Staff Evacuation	High	105	14.3	2026-03-07
Dubai International Airport	DXB	UAE	Middle East	ATC Flow Restriction	Severe	142	14.4	2026-03-04
Cairo International Airport	CAI	Egypt	North Africa	Ground Stop	Low	144	16.4	2026-03-07

join flight_cancellations and airline_losses. Show the airline name, their total passengers_affected, and their estimated_loss_usd

Result Grid   Filter Rows: Export: 

airline	total_passengers_affected	estimated_loss_usd
Emirates	5645	1192640000
Saudi Arabian Airlines	7333	1187560000
Kuwait Airways	7151	1028610000
flydubai	9056	924590000
Mahan Air	6911	903210000
Oman Air	7320	615370000
Qatar Airways	8478	592700000
Emirates SkyCargo	7156	582080000
Gulf Air	7880	421170000
Etihad Airways	10556	324960000
Jazeera Airways	9474	309490000
Iran Air	9220	297950000
Pakistan International	8381	218630000
Air Arabia	7261	190120000
IndiGo	8120	131380000
FedEx Express	9956	102640000
EgyptAir	7797	99760000
Air France	8472	89670000
Singapore Airlines	7480	88000000
Delta Air Lines	10344	80540000

```
3 SELECT
4     l.airline,
5     SUM(c.passengers_affected) AS total_passengers_affected,
6     l.estimated_loss_usd
7 FROM
8     airline_losses l
9     JOIN
10    flight_cancellations c ON l.airline = c.airline
11 GROUP BY l.airline , l.estimated_loss_usd
12 ORDER BY l.estimated_loss_usd DESC;
```


Closure Reasonings: Identify which reason for airspace closure in airspace_closures resulted in the highest total duration_hours.

```
3 SELECT
4     Region, SUM(duration_hours) AS Total_hours
5 FROM
6     airspace_closures
7 GROUP BY region
8 ORDER BY Total_hours desc
9 LIMIT 1;
```

Result Grid   Filter Rows

Region	Total_hours
Middle East	4158

Moving Average: Calculate a 3-day moving average of extra_fuel_cost_usd for 'Qatar Airways' to identify cost spikes.

```
3 SELECT
4     date,
5     daily_fuel_cost,
6     round(AVG(daily_fuel_cost) OVER (ORDER BY date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW),2) AS moving_avg_3day
7 FROM (
8     SELECT
9         date,
10        SUM(extra_fuel_cost_usd) AS daily_fuel_cost
11    FROM
12        flight_reroutes
13    WHERE
14        airline = 'Qatar Airways'
15    GROUP BY
16        date
17 ) AS daily_summaries
18 ORDER BY
19     date;
```

Result Grid   Filter Rows: Search			
	date	daily_fuel_cost	moving_avg_3day
	2026-03-08	9137.32	9137.32
	2026-03-09	4742.26	6939.79
	2026-03-11	13619.03	9166.20
	2026-03-16	11423.28	9928.19
	2026-03-22	6736.97	10593.09
	2026-03-23	3766.71	7308.99
	2026-03-26	6653.43	5719.04
	2026-03-27	9242.97	6554.37
	2026-04-12	4576.40	6824.27
	2026-04-13	8443.62	7421.00
	2026-04-14	7896.71	6972.24
	2026-04-16	13127.21	9822.51
	2026-04-18	3733.69	8252.54



Correlated Subquery: Find airlines in airline_losses that have a revenue_loss_pct higher than the average loss percentage of their entire region.

```
2 SELECT
3     airline, region, revenue_loss_pct
4 FROM
5     airline_losses
6 WHERE
7     revenue_loss_pct > (SELECT
8                         AVG(revenue_loss_pct)
9                         FROM
10                            airline_losses)
11 ORDER BY revenue_loss_pct DESC;
```

Result Grid   Filter Rows: Search			
	airline	region	revenue_loss_pct
	flydubai	Middle East	50.6
	Emirates SkyCargo	Middle East	48.1
	Iran Air	Middle East	44.5
	Air Arabia	Middle East	43.5
	Jazeera Airways	Middle East	40.4
	Etihad Airways	Middle East	34.5
	Oman Air	Middle East	33.5
	Saudi Arabian Airlines	Middle East	33
	Gulf Air	Middle East	28.8
	Mahan Air	Middle East	24.9
	Kuwait Airways	Middle East	24.7
	IndiGo	South Asia	24.5
	Pakistan Internation...	South Asia	24.4
	Qatar Airways	Middle East	23
	Emirates	Middle East	20.5

Lead/Lag Analysis: Use the LAG() function on flight_reroutes to compare the additional_distance_km of a flight to the distance of the previous flight

```
3  SELECT
4      date,
5      airline,
6      flight_number,
7      additional_distance_km,
8      LAG(additional_distance_km) OVER (PARTITION BY airline ORDER BY date) AS prev_flight_distance_km,
9      (additional_distance_km - LAG(additional_distance_km) OVER (PARTITION BY airline ORDER BY date)) AS distance_change_km
10 FROM
11     flight_reroutes
12 ORDER BY
13     airline,
14     date;
```

Result Grid   Filter Rows: Export: 

	date	airline	flight_number	additional_distance_km	prev_flight_distance_km	distance_change_km
	2026-02-28	Air Arabia	AI4779	2742	NULL	NULL
	2026-03-04	Air Arabia	AI2337	2397	2742	-345
	2026-03-04	Air Arabia	AI8904	1786	2397	-611
	2026-03-08	Air Arabia	AI7213	1013	1786	-773
	2026-03-09	Air Arabia	AI131	1721	1013	708
	2026-03-09	Air Arabia	AI3815	2419	1721	698
	2026-03-10	Air Arabia	AI3878	646	2419	-1773
	2026-03-13	Air Arabia	AI7983	1369	646	723
	2026-03-16	Air Arabia	AI5860	925	1369	-444
	2026-03-16	Air Arabia	AI4486	1215	925	290
	2026-03-18	Air Arabia	AI4395	1580	1215	365
	2026-03-20	Air Arabia	AI1178	1612	1580	32
	2026-03-21	Air Arabia	AI6713	1182	1612	-430
	2026-03-23	Air Arabia	AI7730	1350	1182	168
	2026-03-24	Air Arabia	AI3687	1545	1350	195



Complex Analytics (CTE): Create a CTE that calculates the total additional_fuel_cost_usd per country from the estimate table, and then join it with airspace_closures to see if countries with higher costs also had more airspace closures.

Result Grid  Filter Rows: Search			
	country	total_fuel_cost	total_closures
	UAE	4328600	5
	Qatar	1765800	5
	Turkey	1432600	3
	India	916800	2
	Saudi Arabia	648000	3
	Pakistan	520400	1
	Bahrain	424000	2
	Oman	374000	2
	Jordan	272000	4
	Egypt	234000	3
	Israel	0	2

```
4  WITH CountryFuelCosts AS (  
5  
6      SELECT  
7          country,  
8          SUM(additional_fuel_cost_usd) AS total_fuel_cost  
9      FROM  
10         airline_losses_estimate  
11      GROUP BY  
12         country  
13  ),  
14  CountryClosures AS (  
15      SELECT  
16          country,  
17          COUNT(*) AS total_closures  
18      FROM  
19         airspace_closures  
20      GROUP BY  
21         country  
22  )  
23  SELECT  
24      f.country,  
25      f.total_fuel_cost,  
26      c.total_closures  
27  FROM  
28      CountryFuelCosts f  
29  JOIN  
30      CountryClosures c ON f.country = c.country  
31  ORDER BY  
32      f.total_fuel_cost DESC;
```




Thank
You